Spectral Admissibility in the Q_8 Cayley Graph: A Minimal Toy Model for Bounded-Flux Mode Selection

Working note – Cosmochrony programme

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Abstract

We establish the first analytically tractable instance of the spectral admissibility programme outlined in the Cosmochrony framework. On the undirected Cayley graph of the quaternion group Q_8, with generators $S = \{\pm i, \pm j, \pm k\}$, the local bounded-flux constraint $|\partial_t \chi_v| \leq c_\chi$ induces a hierarchy of maximal admissible effective amplitudes across irreducible spectral sectors. In the linearised dispersion regime, the non-abelian irreducible sector admits a maximal effective amplitude larger than the nontrivial abelian sectors by a factor of $\sqrt{4/3} \approx 1.155$. Resolving explicitly the basis dependence in the degenerate ρ_4 eigenspace, we show that this advantage holds for the physically relevant effective amplitude $A_n = a_n \|\psi_n\|_\infty$, but not for the raw amplitude a_n. We then perform the weakly non-linear modal reduction of the discrete DBI Lagrangian and derive the first amplitude-dependent correction to the frequency, controlled by the quartic shape factor $\kappa_n = \sum_v \psi_n(v)^4$. In the canonical Q_8 basis, this correction is softer in the non-abelian sector when expressed in terms of the effective amplitude. This constitutes a minimal model demonstration that bounded substrate capacity may structurally favour non-abelian relational modes, without invoking any additional dynamical selection principle.

1 Setting and Motivation

The Cosmochrony framework identifies the physical spectrum with the residue of two successive constraints acting on the relational operator $\mathcal{L}_\chi$:

1. **Topological discretisation**: the fibre structure of the non-injective projection Π organises modes into discrete winding sectors.
2. **Bounded-flux filtering**: the saturation bound $|\partial_t \chi| \leq c_\chi$ eliminates modes whose local evolution would exceed the substrate's relational capacity.

The goal of this note is to make step 2 explicit and calculable on the simplest non-trivial finite graph encoding a quaternionic algebraic structure, as a minimal discrete proxy for SU(2)-type sectors.

2 The Q_8 Cayley Graph

Let $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ denote the quaternion group of order 8. We define the undirected Cayley graph $\Gamma(Q_8, S)$ with generating set

$$S = \{\pm i, \pm j, \pm k\}. \tag{1}$$

Since S is closed under inversion ($s^{-1} \in S$ for all $s \in S$) and does not contain the identity, $\Gamma(Q_8, S)$ is a well-defined, connected, undirected, 6-regular graph on 8 vertices.

Remark 1. *The choice $S = \{i, j, k\}$ (without inverses) would yield a directed graph with an asymmetric adjacency operator. For a self-adjoint Laplacian, the full symmetric set $S = \{\pm i, \pm j, \pm k\}$ is required. The resulting graph is 6-regular, not 3-regular.*

2.1 Adjacency Operator and Graph Laplacian

The adjacency operator A of $\Gamma(Q_8, S)$ acts on functions $f\colon Q_8 \to \mathbb{C}$ by

$$(Af)(g) = \sum_{s\in S} f(gs) = \sum_{s\in\{\pm i,\pm j,\pm k\}} f(gs). \tag{2}$$

The combinatorial graph Laplacian is

$$\mathcal{L} = dI - A, \qquad d = |S| = 6, \tag{3}$$

where I is the identity on $\ell^2(Q_8)$. The eigenvalues of $\mathcal{L}$ are $\lambda = 6 - \mu$, where μ ranges over the eigenvalues of A.

3 Spectral Decomposition via Representation Theory

Since Q_8 is a finite group, $\ell^2(Q_8) \cong \mathbb{C}[Q_8]$ decomposes as a Q_8-module into irreducible representations. By the Peter–Weyl theorem for finite groups:

$$\mathbb{C}[Q_8] \cong \bigoplus_{\rho\in\widehat{Q_8}} V_\rho^{\oplus \dim\rho}, \tag{4}$$

where the sum runs over the five irreducible representations of Q_8.

3.1 Irreducible Representations of Q_8

Q_8 has five irreducible representations:

Label	Dimension	$\rho(i)$	$\rho(j)$	$\rho(k)$
ρ_0 (trivial)	1	1	1	1
ρ_1	1	1	-1	-1
ρ_2	1	-1	1	-1
ρ_3	1	-1	-1	1
ρ_4 (quaternionic)	2	$\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$

The dimension check: $1^2 + 1^2 + 1^2 + 1^2 + 2^2 = 8 = |Q_8|$. ✓

3.2 Eigenvalues of the Adjacency Operator

For any irreducible representation ρ, the adjacency operator acts as the scalar $\mu_\rho \cdot I_{V_\rho}$ on V_ρ. Since S is a union of conjugacy classes of Q_8, the corresponding element of the group algebra is central, and Schur's lemma gives

$$\mu_\rho = \frac{1}{\dim\rho} \sum_{s\in S} \chi_\rho(s), \tag{5}$$

where χ_ρ is the character of ρ.

1D sectors. For the trivial representation ρ_0: $\chi_{\rho_0}(s) = 1$ for all $s \in S$, so $\sum_{s\in S} \chi_{\rho_0}(s) = 6$ and

$$\mu_{\rho_0} = \frac{1}{1} \cdot 6, \qquad \lambda_{\rho_0} = 6 - 6 = 0. \tag{6}$$

For ρ_1: $\chi_{\rho_1}(\pm i) = 1$ and $\chi_{\rho_1}(\pm j) = \chi_{\rho_1}(\pm k) = -1$, so $\sum_{s\in S} \chi_{\rho_1}(s) = 2(1) + 4(-1) = -2$, giving $\mu_{\rho_1} = -2$ and $\lambda_{\rho_1} = 8$. By the symmetry of Q_8 under permutations of $\{i, j, k\}$, $\mu_{\rho_2} = \mu_{\rho_3} = -2$ and $\lambda_{\rho_2} = \lambda_{\rho_3} = 8$.

2D sector. For the quaternionic representation ρ_4, we compute the character at each generator. The matrices $\rho_4(i)$, $\rho_4(-i)$, $\rho_4(j)$, $\rho_4(-j)$, $\rho_4(k)$, $\rho_4(-k)$ each have trace zero: for instance, $\operatorname{tr}(\rho_4(i)) = \operatorname{tr}\left(\begin{smallmatrix} i & 0 \\ 0 & -i \end{smallmatrix}\right) = 0$, and $\operatorname{tr}(\rho_4(-i)) = 0$; similarly for j and k. Thus $\sum_{s \in S} \chi_{\rho_4}(s) = 0$ and

$$\mu_{\rho_4} = \frac{1}{2} \cdot 0 = 0, \qquad \lambda_{\rho_4} = 6 - 0 = 6. \tag{7}$$

3.3 Summary of the Spectrum

Sector	λ_n	Multiplicity	Type
ρ_0	0	1	Trivial (zero mode)
ρ_1, ρ_2, ρ_3	8	3	Nontrivial abelian
ρ_4	6	4	Non-abelian (quaternionic)

Total dimension: $1 + 3 + 4 = 8$. ✓

4 Bounded-Flux Admissibility Condition

4.1 Setup

Let the substrate field $\chi \colon Q_8 \times \mathbb{R} \to \mathbb{R}$ evolve on the graph, subject to the local bounded-flux constraint

$$|\partial_t \chi_v(t)| \leq c_\chi \qquad \forall v \in Q_8,\ \forall t. \tag{8}$$

Consider a single-mode excitation

$$\chi_v(t) = a_n \psi_n(v) \cos(\omega_n t), \tag{9}$$

where ψ_n is an eigenmode of $\mathcal{L}$ with eigenvalue λ_n, normalised in $\ell^2(Q_8)$, and $a_n > 0$ is the mode amplitude. The time derivative at vertex v is

$$\partial_t \chi_v(t) = -a_n \omega_n \psi_n(v) \sin(\omega_n t). \tag{10}$$

The constraint (8) requires

$$|\omega_n| \cdot a_n \cdot |\psi_n(v)| \leq c_\chi \qquad \forall v \in Q_8. \tag{11}$$

The binding constraint is at the vertex where $|\psi_n(v)|$ is maximal:

$$|\omega_n| \cdot a_n \cdot \|\psi_n\|_\infty \leq c_\chi. \tag{12}$$

4.2 Effective Amplitude and Admissibility Envelope

Definition 1. *The* effective amplitude *of a mode* (ψ_n, a_n) *is*

$$A_n := a_n \|\psi_n\|_\infty. \tag{13}$$

A mode is dynamically admissible *(in the linearised regime) if*

$$|\omega_n| \cdot A_n \leq c_\chi. \tag{14}$$

In the linearised dispersion approximation, $\omega_n^2 \approx \lambda_n$, so the maximal admissible effective amplitude is

$$A_n^{\max} = \frac{c_\chi}{\sqrt{\lambda_n}}. \tag{15}$$

This defines the *admissibility envelope* in the (λ_n, A_n) plane.

Remark 2. *The linearised dispersion $\omega_n^2 = \lambda_n$ is valid for $A_n \ll c_\chi/\sqrt{\lambda_n}$. The non-linear (Born-Infeld) correction modifies this relation near saturation and requires a modal reduction of the DBI action; it is deferred to a separate calculation. The envelope* (15) *should therefore be understood as the admissibility boundary in the weakly non-linear regime.*

4.3 Explicit Eigenmodes in the ρ_4 Sector

The non-abelian sector ρ_4 has multiplicity 4 in $\ell^2(Q_8)$, so the quantity $\|\psi_n\|_\infty$ is not determined until an explicit basis of the eigenspace is chosen. A natural orthonormal basis is provided by the matrix coefficients of the quaternionic irreducible representation.

For a finite group G and an irreducible representation ρ, the standard Peter–Weyl basis of the corresponding isotypic component is given by

$$\psi_{uv}(g) = \sqrt{\frac{\dim \rho}{|G|}}\, \rho(g)_{uv}, \qquad u, v \in \{1, \dots, \dim \rho\}. \tag{16}$$

For Q_8 and ρ_4, this becomes

$$\psi_{uv}(g) = \frac{1}{2}\, \rho_4(g)_{uv}, \qquad u, v \in \{1, 2\}, \tag{17}$$

since $\dim \rho_4 = 2$ and $|Q_8| = 8$.

Using the explicit matrices

$$\rho_4(1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \rho_4(-1) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{18}$$

$$\rho_4(i) = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad \rho_4(j) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \rho_4(k) = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \tag{19}$$

one finds that the diagonal modes ψ_{11}, ψ_{22} are supported on $\{\pm 1, \pm i\}$, while the off-diagonal modes ψ_{12}, ψ_{21} are supported on $\{\pm j, \pm k\}$. In all four cases,

$$|\psi_{uv}(g)| \in \left\{0, \frac{1}{2}\right\}, \tag{20}$$

hence

$$\|\psi_{uv}\|_\infty = \frac{1}{2}, \qquad u, v \in \{1, 2\}. \tag{21}$$

For the one-dimensional nontrivial abelian sectors ρ_1, ρ_2, ρ_3, the normalised eigenmodes are the corresponding characters scaled by $|Q_8|^{-1/2}$. Since these characters take values in $\{\pm 1\}$, one has

$$\|\psi_{\rho_k}\|_\infty = \frac{1}{\sqrt{8}}, \qquad k = 1, 2, 3. \tag{22}$$

4.4 Basis-Resolved Admissibility

Proposition 1 (Basis-resolved admissibility on Q_8)**.** *In the canonical Peter–Weyl basis of $\ell^2(Q_8)$, the maximal admissible effective amplitudes satisfy*

$$A_{\rho_4}^{\max} = \frac{c_\chi}{\sqrt{6}} > \frac{c_\chi}{\sqrt{8}} = A_{\rho_k}^{\max}, \qquad k = 1, 2, 3, \tag{23}$$

whereas the corresponding maximal raw amplitudes satisfy

$$a_{\rho_4}^{\max} = \frac{c_\chi}{\sqrt{6}\, \|\psi_{\rho_4}\|_\infty} = \frac{2}{\sqrt{6}}\, c_\chi \approx 0.816\, c_\chi, \tag{24}$$

$$a_{\rho_k}^{\max} = \frac{c_\chi}{\sqrt{8}\, \|\psi_{\rho_k}\|_\infty} = c_\chi, \qquad k = 1, 2, 3. \tag{25}$$

Thus the non-abelian sector is favoured in effective amplitude, but not in raw amplitude.

Proof. The effective-amplitude statement follows from $A_n^{\max} = c_\chi/\sqrt{\lambda_n}$ together with $\lambda_{\rho_4} = 6$ and $\lambda_{\rho_k} = 8$. The raw-amplitude statement follows from $a_n^{\max} = A_n^{\max}/\|\psi_n\|_\infty$, using $\|\psi_{\rho_4}\|_\infty = 1/2$ and $\|\psi_{\rho_k}\|_\infty = 1/\sqrt{8}$. □

Remark 3 (Interpretation)**.** *The bounded-flux constraint acts on the local quantity $|\omega_n|\, a_n\, |\psi_n(v)|$, and therefore selects modes through the effective amplitude $A_n = a_n\|\psi_n\|_\infty$, not through the raw amplitude a_n alone. The non-abelian advantage on Q_8 is therefore not a larger absolute amplitude capacity, but a larger admissible effective window per unit spectral weight. Equivalently, the ρ_4 sector is less constrained by bounded-flux filtering in the physically relevant local variable.*

4.5 Weakly Non-linear Modal DBI Reduction

To incorporate the first Born–Infeld correction, consider the discrete DBI Lagrangian

$$L = -c_\chi^2 \sum_v \sqrt{1 - \dot{\chi}_v^2/c_\chi^2} - \tfrac{1}{2} \sum_{v,w} \chi_v \mathcal{L}_{vw} \chi_w. \tag{26}$$

For a single eigenmode,

$$\chi_v(t) = q(t)\psi_n(v), \tag{27}$$

with $\mathcal{L}\psi_n = \lambda_n\psi_n$ and $\|\psi_n\|_2 = 1$, this reduces to the exact one-mode Lagrangian

$$L_n[q] = -c_\chi^2 \sum_v \sqrt{1 - \psi_n(v)^2 \dot{q}^{\,2}/c_\chi^2} - \tfrac{1}{2}\lambda_n q^2. \tag{28}$$

Expanding the square root for $|\dot{q}| \ll c_\chi$ gives

$$L_n[q] = \text{const} + \tfrac{1}{2}\dot{q}^{\,2} + \frac{\kappa_n}{8c_\chi^2}\dot{q}^{\,4} - \tfrac{1}{2}\lambda_n q^2 + \mathcal{O}(c_\chi^{-4}), \tag{29}$$

where

$$\kappa_n := \sum_v \psi_n(v)^4. \tag{30}$$

The corresponding Euler–Lagrange equation is

$$\left(1 + \frac{3\kappa_n}{2c_\chi^2}\dot{q}^{\,2}\right)\ddot{q} + \lambda_n q = 0. \tag{31}$$

Using the harmonic-balance ansatz

$$q(t) \approx a_n \cos(\omega_n t), \tag{32}$$

and keeping only the fundamental harmonic, one obtains the first amplitude-dependent correction

$$\omega_n^2(a_n) \approx \lambda_n \left(1 - \frac{3\kappa_n\lambda_n a_n^2}{8c_\chi^2}\right), \tag{33}$$

or equivalently

$$\omega_n(a_n) \approx \sqrt{\lambda_n}\left(1 - \frac{3\kappa_n\lambda_n a_n^2}{16c_\chi^2}\right). \tag{34}$$

The DBI correction is therefore a softening non-linearity: the frequency decreases as the mode amplitude grows.

In the canonical Q_8 basis, the quartic shape factor is explicit. For the one-dimensional nontrivial abelian sectors,

$$\kappa_{\rho_k} = \sum_{v\in Q_8}\left(\frac{1}{\sqrt{8}}\right)^4 = \frac{1}{8}, \qquad k = 1, 2, 3. \tag{35}$$

For each canonical basis vector in the ρ_4 sector,

$$\kappa_{\rho_4} = 4\left(\frac{1}{2}\right)^4 = \frac{1}{4}. \tag{36}$$

Hence the raw-amplitude correction is stronger in the non-abelian sector, reflecting the larger localisation of the corresponding basis vectors.

However, rewriting the correction in terms of the effective amplitude $A_n = a_n\|\psi_n\|_\infty$ gives

$$\omega_n^2(A_n) \approx \lambda_n\left(1 - \frac{3\lambda_n A_n^2}{8c_\chi^2}\frac{\kappa_n}{\|\psi_n\|_\infty^2}\right). \tag{37}$$

In the canonical Q_8 basis,

$$\frac{\kappa_{\rho_k}}{\|\psi_{\rho_k}\|_\infty^2} = \frac{(1/8)}{(1/\sqrt{8})^2} = 1, \qquad \frac{\kappa_{\rho_4}}{\|\psi_{\rho_4}\|_\infty^2} = \frac{(1/4)}{(1/2)^2} = 1. \tag{38}$$

Therefore

$$\omega_n^2(A_n) \approx \lambda_n\left(1 - \frac{3\lambda_n A_n^2}{8c_\chi^2}\right) \tag{39}$$

in both the nontrivial abelian and non-abelian sectors. At fixed effective amplitude, the non-abelian sector thus remains less penalised by the weakly non-linear DBI correction, solely because $\lambda_{\rho_4} = 6$ is smaller than $\lambda_{\rho_k} = 8$.

4.6 Non-perturbative DBI envelope

The weakly non-linear reduction derived above provides the first correction to the dispersion relation. However, the Dirac–Born–Infeld dynamics possesses an exact single-mode description that allows the admissibility boundary to be determined non-perturbatively.

Consider again a single eigenmode

$$\chi_v(t) = q(t)\psi_n(v)$$

with $\mathcal{L}\psi_n = \lambda_n\psi_n$.

Substituting into the discrete DBI Lagrangian yields the exact one-mode Lagrangian

$$L(q,\dot{q}) = -c_\chi^2 \sum_{v\in Q_8}\sqrt{1 - \frac{\dot{q}^2\psi_n(v)^2}{c_\chi^2}} - \frac{1}{2}\lambda_n q^2. \tag{40}$$

Energy conservation

Since the Lagrangian is time independent, the corresponding Hamiltonian is conserved. One obtains

$$E = c_\chi^2 \sum_{v\in Q_8}\frac{1}{\sqrt{1 - \frac{\dot{q}^2\psi_n(v)^2}{c_\chi^2}}} + \frac{1}{2}\lambda_n q^2. \tag{41}$$

At turning points $\dot{q} = 0$ the energy reduces to

$$E = 8c_\chi^2 + \frac{1}{2}\lambda_n A_n^2, \tag{42}$$

where A_n denotes the oscillation amplitude.

Velocity bound

The DBI square root imposes the constraint

$$|\dot{q}|\,|\psi_n(v)| < c_\chi \qquad \forall v.$$

The most restrictive condition is obtained at the vertex where $|\psi_n|$ is maximal, yielding

$$|\dot{q}| < \frac{c_\chi}{\|\psi_n\|_\infty}. \tag{43}$$

Thus the modal coordinate $q(t)$ obeys a relativistic-type speed limit.

Period–amplitude relation

Combining the energy conservation law with (**??**) gives

$$\dot{q} = \frac{c_\chi}{\|\psi_n\|_\infty}\sqrt{1 - \left(\frac{\lambda_n q^2}{\lambda_n A_n^2}\right)}.$$

The oscillation period therefore becomes

$$T(A_n) = 4\int_0^{A_n} \frac{dq}{\dot{q}}. \tag{44}$$

Near the saturation limit

$$A_n \to A_n^{\max},$$

the velocity approaches the DBI bound and the integral develops a logarithmic divergence.

Consequently the oscillation frequency behaves as

$$\omega_n(A_n) \to 0 \qquad (A_n \to A_n^{\max}). \tag{45}$$

Interpretation

The non-perturbative DBI dynamics therefore produces a dynamical freezing of modes approaching the saturation boundary. Modes close to the admissibility envelope oscillate increasingly slowly, effectively trapping energy near the lowest admissible spectral sectors.

This mechanism strengthens the spectral filtering effect identified in the linear analysis. Modes with larger eigenvalues λ_n approach the freezing regime at smaller effective amplitudes, reinforcing the advantage of the lowest spectral sector.

5 Main Result

Proposition 2 (Spectral admissibility hierarchy on Q_8)**.** *In the undirected Cayley graph $\Gamma(Q_8, \{\pm i, \pm j, \pm k\})$, under the local bounded-flux constraint* (8) *and linearised dispersion, the maximal admissible effective amplitude satisfies*

$$A_{\rho_4}^{\max} = \frac{c_\chi}{\sqrt{6}} > \frac{c_\chi}{\sqrt{8}} = A_{\rho_k}^{\max}, \quad k = 1, 2, 3. \tag{46}$$

The non-abelian sector ρ_4 occupies a strictly larger admissible region in the space of effective mode amplitudes than any nontrivial abelian sector, by a factor

$$\frac{A_{\rho_4}^{\max}}{A_{\rho_k}^{\max}} = \sqrt{\frac{8}{6}} = \sqrt{\frac{4}{3}} \approx 1.155. \tag{47}$$

Proof. Direct substitution of $\lambda_{\rho_4} = 6$ and $\lambda_{\rho_k} = 8$ into the admissibility envelope (15). □

Remark 4. *The trivial sector ρ_0 has $\lambda = 0$ and thus formally unbounded $A^{\max}$. It corresponds to the uniform zero mode of the connected graph. In the present analysis it is factored out, since we are interested only in non-trivial excitations above the connected background.*

6 Interpretation

Proposition 2, together with the basis-resolved Proposition 1, establishes that, at fixed substrate capacity c_χ, the non-abelian irreducible sector of the quaternion Cayley graph sustains a larger local effective amplitude before violating the bounded-flux constraint, even though its raw amplitude bound is lower than that of the nontrivial abelian sectors.

In the language of the Cosmochrony programme:

- The admissibility envelope $A_n^{\max} = c_\chi/\sqrt{\lambda_n}$ defines a *geometry of exclusion* in the space of relational modes: modes above the envelope are dynamically inadmissible.
- The non-abelian sector ρ_4 lies strictly below this envelope for a larger range of effective amplitudes than the nontrivial abelian sectors ρ_1, ρ_2, ρ_3: the bounded-flux constraint is less restrictive for non-abelian modes, enlarging their admissible window by a factor of $\sqrt{4/3}$.
- This advantage is a statement about the physically relevant local variable $A_n = a_n\|\psi_n\|_\infty$, not about the raw amplitude a_n. In the canonical basis, the ρ_4 modes are more localised ($\|\psi\|_\infty = 1/2$) than the abelian ones ($\|\psi\|_\infty = 1/\sqrt{8}$), so the non-abelian sector is more constrained in raw amplitude even while remaining less constrained in effective amplitude.
- The weakly non-linear DBI correction is governed by the quartic shape factor $\kappa_n = \sum_v \psi_n(v)^4$. In raw amplitude, the canonical ρ_4 modes are more strongly penalised than the abelian ones because they are more localised. In effective amplitude, this localisation factor cancels on Q_8, so the first DBI correction again favours the non-abelian sector through its smaller eigenvalue.
- This provides a minimal admissibility advantage for non-abelian relational modes, both at the linear level and at the first weakly non-linear DBI order, without any dynamical postulate beyond the local flux bound.

Remark 5 (Real field and complex eigenmodes)**.** *The substrate field χ is taken real-valued. The natural eigenmodes of the 2D sector are matrix coefficients of ρ_4, which are complex-valued. One may either work in $\ell^2(Q_8, \mathbb{C})$ and take real and imaginary parts to obtain real eigenmodes, or directly choose a real basis for each eigenspace. The local admissibility criterion is intrinsically defined, but its expression in terms of (a_n, ψ_n) depends on the chosen basis within a degenerate eigenspace through $\|\psi_n\|_\infty$. This basis dependence is resolved explicitly in Section 4.3.*

This result does not yet establish:

- that the non-abelian sector will dominate in a full dynamical simulation;
- the origin of the eigenvalue gap $\lambda_{\rho_4} = 6 < 8 = \lambda_{\rho_k}$ from the topological structure of Π alone;
- the fully non-linear Born–Infeld correction to the admissibility envelope near saturation beyond the weakly non-linear regime.

These remain open and constitute the next steps of the programme.

7 Synthesis: Spectral Filtering by Bounded Flux

The analysis performed in this note identifies the effective amplitude

$$A_n = a_n \|\psi_n\|_\infty$$

as the physically relevant variable controlling the saturation of the substrate.

7.1 Local Flux Constraint as a Spectral Filter

The bounded-flux condition

$$|\partial_t \chi_v| \leq c_\chi$$

acts locally on each vertex of the relational substrate.

For a single eigenmode

$$\chi_v(t) = a_n \psi_n(v) \cos(\omega_n t),$$

the constraint becomes

$$|\omega_n|\, a_n\, \|\psi_n\|_\infty \leq c_\chi.$$

Introducing the effective amplitude

$$A_n = a_n \|\psi_n\|_\infty$$

yields the admissibility envelope

$$A_n^{\max} = \frac{c_\chi}{\sqrt{\lambda_n}}.$$

A key consequence is that the localization of the mode does not alter the admissibility bound itself. Localization only determines how the raw amplitude a_n translates into a local flux impact on the substrate. The selection mechanism is therefore governed purely by the spectral eigenvalue λ_n.

7.2 DBI Softening as Dynamical Spectral Filtering

In the weakly non-linear regime the discrete Dirac–Born–Infeld dynamics produces an amplitude-dependent frequency

$$\omega_n^2(A_n) \approx \lambda_n \left(1 - \frac{3\lambda_n A_n^2}{8c_\chi^2}\right).$$

This simplified expression follows from the identity

$$\frac{\kappa_n}{\|\psi_n\|_\infty^2} = 1$$

which holds for all non-trivial sectors of the Q_8 Cayley graph.

The non-linear correction therefore depends only on the Laplacian eigenvalue. Modes with larger λ_n undergo stronger softening as the effective amplitude increases.

The relational substrate thus behaves as a dynamical low-pass spectral filter.

7.3 Conceptual Interpretation

The mechanism uncovered in this toy model can be summarized as

$$\text{substrate capacity} \rightarrow \text{spectral admissibility} \rightarrow \text{nonlinear filtering} \rightarrow \text{surviving sectors.}$$

On the Q_8 Cayley graph the smallest non-trivial eigenvalue

$$\lambda_{\rho_4} = 6$$

belongs to the quaternionic non-abelian sector. This sector therefore possesses the largest admissible amplitude window and is the least affected by DBI softening.

In this sense, physical structures correspond to the spectral components that remain dynamically admissible under the bounded capacity of the relational substrate.

Matter may thus be interpreted as the dynamical residue of the spectral filtering imposed by the substrate.

7.4 Beyond the Q_8 Toy Model

The identity

$$\frac{\kappa_n}{\|\psi_n\|_\infty^2} = 1$$

is a special feature of the small and highly symmetric Q_8 graph.

On larger graphs this ratio will generally depend on the structure of the eigenmodes. The non-linear DBI correction may therefore amplify the spectral selection mechanism rather than acting uniformly across sectors.

This motivates extending the analysis to larger Cayley graphs, including Ramanujan families such as the Lubotzky–Phillips–Sarnak constructions on $\mathrm{SL}(2, \mathbb{F}_p)$.

7.5 Next Step: Dynamical Survival

The next stage of the programme is to test this mechanism dynamically.

The proposed protocol is to initialise the substrate with a mixture of spectral modes close to the admissibility envelope and evolve the system under the discrete DBI dynamics.

If spectral cascades preferentially accumulate in the lowest admissible sector, this would demonstrate a genuine dynamical selection of symmetry sectors driven by the bounded relational capacity of the substrate.

8 Next Steps

The present note completes the first two stages of the spectral admissibility programme on the Q_8 Cayley graph.

Step 1: spectral admissibility. The bounded-flux constraint $|\partial_t \chi_v| \leq c_\chi$ selects admissible modes through the effective amplitude $A_n = a_n \|\psi_n\|_\infty$, leading to the envelope

$$A_n^{\max} = \frac{c_\chi}{\sqrt{\lambda_n}}.$$

On Q_8, this produces a strict hierarchy in which the quaternionic non-abelian sector ρ_4 admits the largest effective amplitude window.

Step 2: weakly non-linear DBI reduction. The discrete Dirac–Born–Infeld dynamics has been reduced to a single-mode effective equation, yielding the amplitude-dependent frequency

$$\omega_n^2(A_n) \approx \lambda_n \left(1 - \frac{3\lambda_n A_n^2}{8c_\chi^2}\right).$$

On the Q_8 Cayley graph the ratio $\kappa_n/\|\psi_n\|_\infty^2 = 1$ for all non-trivial sectors, implying that the non-linear softening depends only on the Laplacian eigenvalue. The spectral hierarchy obtained in Step 1 therefore remains stable under the leading DBI correction.

Step 3: non-perturbative DBI envelope. The next task is to move beyond the perturbative expansion in c_χ^{-2} and derive the full amplitude-dependent period from the exact one-mode Lagrangian (28). This can be obtained either through elliptic-integral methods or direct numerical quadrature. The resulting period–amplitude relation will determine the exact non-perturbative admissibility envelope replacing the linear boundary (15).

Step 4: dynamical survival of spectral sectors. The subsequent step is to simulate the full Q_8 substrate with the DBI constraint imposed locally at each vertex. The modal decomposition

$$A_n(t) = a_n(t)\|\psi_n\|_\infty$$

can then be tracked dynamically. The key question is whether spectral cascades concentrate energy in the sector with the largest admissible window, namely the non-abelian ρ_4 sector.

Step 5: extension to larger graphs. Finally, the toy model should be extended to Cayley graphs of larger groups approximating S^3 more faithfully. Natural candidates include the binary icosahedral group $2I$ (order 120) and families of Cayley graphs on $\mathrm{SL}(2, \mathbb{F}_p)$.

In particular, Ramanujan graph constructions such as the Lubotzky–Phillips–Sarnak graphs provide sequences with optimal spectral gap properties. Studying bounded-flux filtering on such graphs would test whether the spectral admissibility hierarchy persists in large expanders and may reveal a deeper connection between substrate capacity constraints and the spectral theory of expander graphs.